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Statistical analysis of textual data

- Text as data
- Text representations

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Today's goals

- Text as data
- Text representations

1. What makes text data different?
2. How do (textual) representations create measurements?
3. Why do modeling choices matter for inference?



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- Text as data
- Text representations

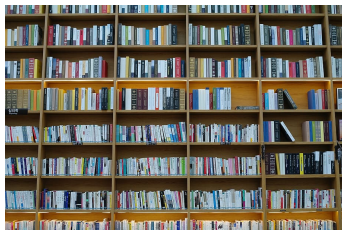
Section 1

Text as data



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Why?

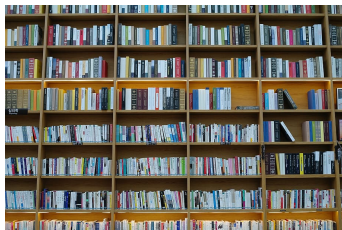


- Large textual corpora are (or are becoming) digital



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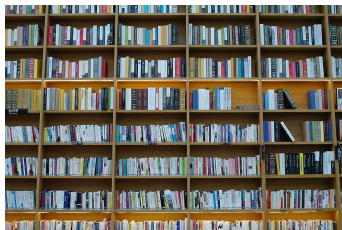


- Text as data
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- Large textual corpora are (or are becoming) digital
 - Why now?



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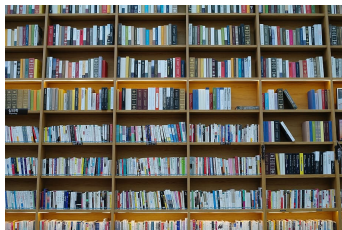
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- Text as data
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- Large textual corpora are (or are becoming) digital
- Why now? Digitization, archives/APIs, computational power, transformer models



Why?



- Text as data
 - Text representations
- Large textual corpora are (or are becoming) digital
 - Why now? Digitization, archives/APIs, computational power, transformer models
 - We want to study **meaning** and **semantics** to draw conclusions about society: treating **text as data**.
 - Interest in **inference** from data, **predictions** or/and a combination.



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Examples of problems

- Text as data
- Text representations

Research question	Text data	Target quantity	Method family
How does the meaning of "propaganda" change?	Books, newspapers	Semantic change	Word embeddings



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How do public discourses on immigration change?	Newspapers, debates, policy text	Topics, frames, prevalence	Topic models



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What affects the outcome of a court case?	Decisions, briefs, metadata	Labels, decisions, risk	Supervised models



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Why language is hard

The quick brown fox jumps over the lazy dog.

- Text as data
- Text representations
- Text has structure at many levels



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 - characters,



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 - syntax,



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 - syntax, context , genre , pragmatics (social dimension of language)



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Why inference from text is hard

- The scientific target is often **social**, not textual
 - public opinion,

- Text as data
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- We know that all models are wrong.
But **some models can be useful**.



How to measure and estimate meaning

Closing arguments were heard yesterday in the Federal bankruptcy fraud trial of Stephen J. Sabbath, whose legal problems have raised doubts about his ability to continue as leader of the Nassau County Democratic Party.

- The distributional hypothesis (see Sahlgren, 2008)
a word is characterized by the company it keeps
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 - left context (predict the next word)



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From statistical models to LLMs

- **1980s–90s: Statistical models and n-grams**

Text as counts, local word sequences, and sparse matrices;
useful for simple classification and search.

- Text as data
- Text representations



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- **2017–21: Transformers and encoders/BERT**
Attention Is All You Need: contextual representations from attention over tokens.
- **2022–: Instruction-tuned LLMs**
Prompt-conditioned generative models; useful for annotation, summarization, coding assistance, and structured extraction.

The main shift: from counting words to modeling context and generated language.



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The most common model classes

Model class	Uses	Typical outputs
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Encoder models	bidirectional context and masks	document labels, contextual embeddings, extractions
Decoder models/LLMs	left context and next-word prediction	generated text, annotations, structured extractions



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How should we do it? The Box process

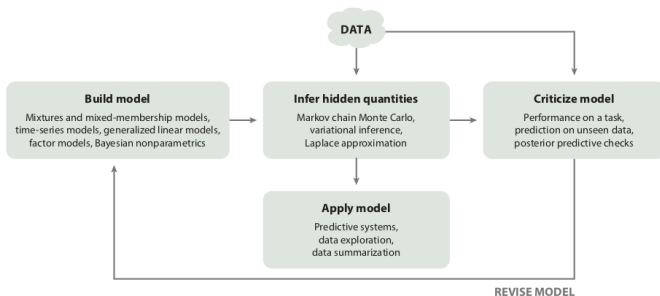


Figure: The Box approach (Box, 1976, Blei, 2014)



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General model checking for text

- Text as data
- Text representations
- Inspect top words, labels, and representative documents



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- Read examples where the model is **confident** and **uncertain**



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- Compare to human annotations or expert-coded benchmarks (e.g. by yourself)



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- Ask whether errors are random, systematic, or substantively meaningful



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Validation and evaluation

- Text as data
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- Central question: does the representation/model **capture the construct of interest?**



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- Substantive validation: expert judgment, hand-coded samples, known cases



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- Design validation: sampling, metadata, time periods, genre, missingness



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- Central question: does the representation/model **capture the construct of interest**?
- Technical evaluation: prediction accuracy, held-out likelihood, calibration
- Substantive validation: expert judgment, hand-coded samples, known cases
- Design validation: sampling, metadata, time periods, genre, missingness
- For social science: **substantive validity** comes before technical validity



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- Text as data
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Section 2

Text representations



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- Text as data
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Basic units in text data

Token

One observed occurrence in a text, e.g. the second "the" in a sentence



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Basic units in text data

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One observed occurrence in a text, e.g. the second "the" in a sentence

Type

A unique vocabulary item, e.g. the word type "the"



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Basic units in text data

Token	One observed occurrence in a text, e.g. the second "the" in a sentence
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Corpus	A collection of documents chosen for a research purpose
Metadata	Information about documents: date, source, speaker, party, genre, outcome



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Representation is measurement

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- We choose units: token, sentence, document, author, time period
- We choose features: counts, embeddings, labels, prompts, metadata



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Representation is measurement

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- We choose transformations: tokenization, normalization, aggregation



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- Text is not naturally a matrix
- We choose units: token, sentence, document, author, time period
- We choose features: counts, embeddings, labels, prompts, metadata
- We choose transformations: tokenization, normalization, aggregation
- Every pipeline choice defines what can and cannot be measured



Computational text representation

- The tidy data format
 - Each variable is a column
 - Each observation is a row
 - Each type of observational unit is a table
- For text: a table with one-token-per-row
- A token is a discrete unit of interest

Example:

The quick brown fox jumps over the lazy dog.

<i>pos</i>	<i>word_type</i>	<i>sentence</i>
1	<i>The</i>	1
2	<i>quick</i>	1
3	<i>brown</i>	1
...	...	...



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Other representations and aggregations

- Other representations
 - sentence/token/character embeddings

- Text as data
- Text representations



Other representations and aggregations

- Other representations
 - sentence/token/character embeddings
- Aggregating text
 - document-term matrices
 - term-term matrices
- Commonly used in simple text classification

Example:

The quick brown fox jumps over the lazy dog. That was quick! ...

	sentence1	sentence2	...
<i>the</i>	2	0	...
<i>quick</i>	1	1	...
<i>brown</i>	1	0	...
...	...	...	...



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What is lost in bag-of-words?

- Text as data
- Text representations
- Word order and syntax



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- Negation, modality, and long-distance dependencies



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- Word order and syntax
 - Negation, modality, and long-distance dependencies
 - Speaker, genre, and context unless attached as metadata



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- Word order and syntax
- Negation, modality, and long-distance dependencies
- Speaker, genre, and context unless attached as metadata
- Document structure, sequence, and interaction
- Irony, sarcasm, quotation, and pragmatic meaning

Useful summaries, but never neutral representations.



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Aggregation using tidy text

The quick brown fox jumps over the lazy dog. That was quick! ...

<i>word_type</i>	<i>sentence</i>	<i>n</i>
<i>the</i>	1	2
<i>quick</i>	1	1
<i>brown</i>	1	1
<i>that</i>	2	1
<i>quick</i>	2	1
...	...	...



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Word tokenization problems

The quick brown fox jumps over the 13 lazy dogs.

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- Simple tokenization: word-based



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- Simple tokenization: word-based
- Difficult in some languages (e.g. Chinese)



Word tokenization problems

The quick brown fox jumps over the 13 lazy dogs.

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 - The "right" tokenization depends on the research question



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Similar to BPE, but uses a language model to score which pairs to merge
- **Unigram tokenizer:**
Unlike BPE/WP, starts with a large vocabulary and trims it down



Practical (vocabulary) curation

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- Text as data
- Lemmatizing/stemming (text normalization)

*The quick brown fox **jump** over the 13 lazy **dog**.*



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***the** quick brown fox jump over the 13 lazy dog.*



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- Normalize numbers and remove punctuation

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*the quick brown fox jump over the **NN** lazy dog*
- Stop words and rare words (Zipf's law, frequency vs. rank)
quick brown fox jump NN lazy dog



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- Stop words and rare words (Zipf's law, frequency vs. rank)
*quick brown fox jump **NN** lazy dog*
- Nota bene! These choices are mostly for count-based/bag-of-words workflows, and are often inappropriate before transformer/LLM models.



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Bridge to the rest of the course

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- Today: representations turn text into analyzable data



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Takeaways

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Takeaways

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- Different models use different notions of context
- Evaluation must connect model output to the social-science construct
- Useful text analysis combines computation, model checking, and close reading